Systems Analysis and Design

IS1208

Rupika Wijesinghe

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

References

1. Alan Dennis, Barbara Haley, David Tegarden, Systems analysis design, An Object Oriented Approach with UML : an object oriented approach, 5th edition, John Wiley & Sons, 2015, ISBN 978-1-118-80467-4
2. Ian Somerville, Software Engineering, 10th edition,Pearson , Pearson Education, 2016, ISBN: 978-0133943030
3. Jeffrey L Whitten & Lonnie D Bentley, System Analysis and DESIGN METHODS, 7th Edition, 2007, ISBN 0-07-063417-3
4. http://www.agilemodeling.com/
5. https://www.visualparadigm.com/editions/community/

Course information

- Course template - refer LMS
  - Learning objectives
  - References
  - Evaluation

Tools and Other Resources

Tools :

https://www.visual-paradigm.com/editions/community/

https://argouml.en.softonic.com/

http://staruml.io/ or any tool that supports UML 2.0

Other Resources:

UML 2.0 Tutorial –

https://www.youtube.com/playlist?list=PLGLfVvz_LVvQ5G-LdJ8RLqendo7QITYc

- Agile Modelling - http://www.agilemodeling.com/

Systems Analysis and Design

Late 80's -
Boring Subject

Interesting after Software tools were introduced (Early 90's in SL)

What is Systems Analysis & Design? Why do you want Systems A & D?

- A Simple Program.

Programming example

```
INPUT X
INPUT Y
ADD=X+Y
PRINT ADD
```

A Library System

Tasks

Lending
Reserving
Returning
Register Members
Register Books

Data

Member
Book
Copy
LendingDetails
ReservationDetails

Information Systems

- **Earlier applications**

Airline Reservation systems

Transaction Processing systems
Keeping records of transactions

Stock control systems
Keeping records of Stock

Information Systems

- Computers are now becoming part of virtually every activity in an organization

Production

HRM - Training

Telephone Integration

Information System…

Information System

- It is an arrangement of

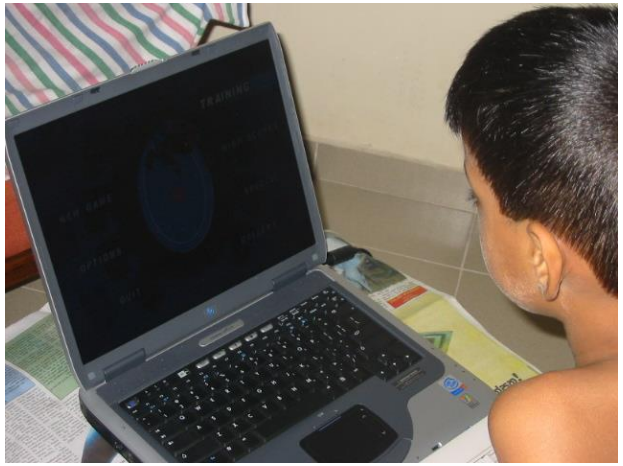
- To support and Improve day to day operations in a Business,
- To support the problem solving and decision-making needs of management and users

Stakeholder is any person who has an interest in an information system and its outputs.

Technical **Non Technical**

System Development Stakeholders

- Users



who will use or is affected by an information system on a regular basis – capturing, validating, entering, responding to, storing, and exchanging data and information.

System Users or Clients

External, Internal

Stakeholders

System Owner

- Owns the system.
- They pay for the system to be built and maintained.
- Set the vision and priorities.
- Determine the policies.
- They also may be system users.

Stakeholders

Systems Analysts

People who understand both business and computing.

They study business problems and opportunities.

Transform business and information requirements into specifications for IS.

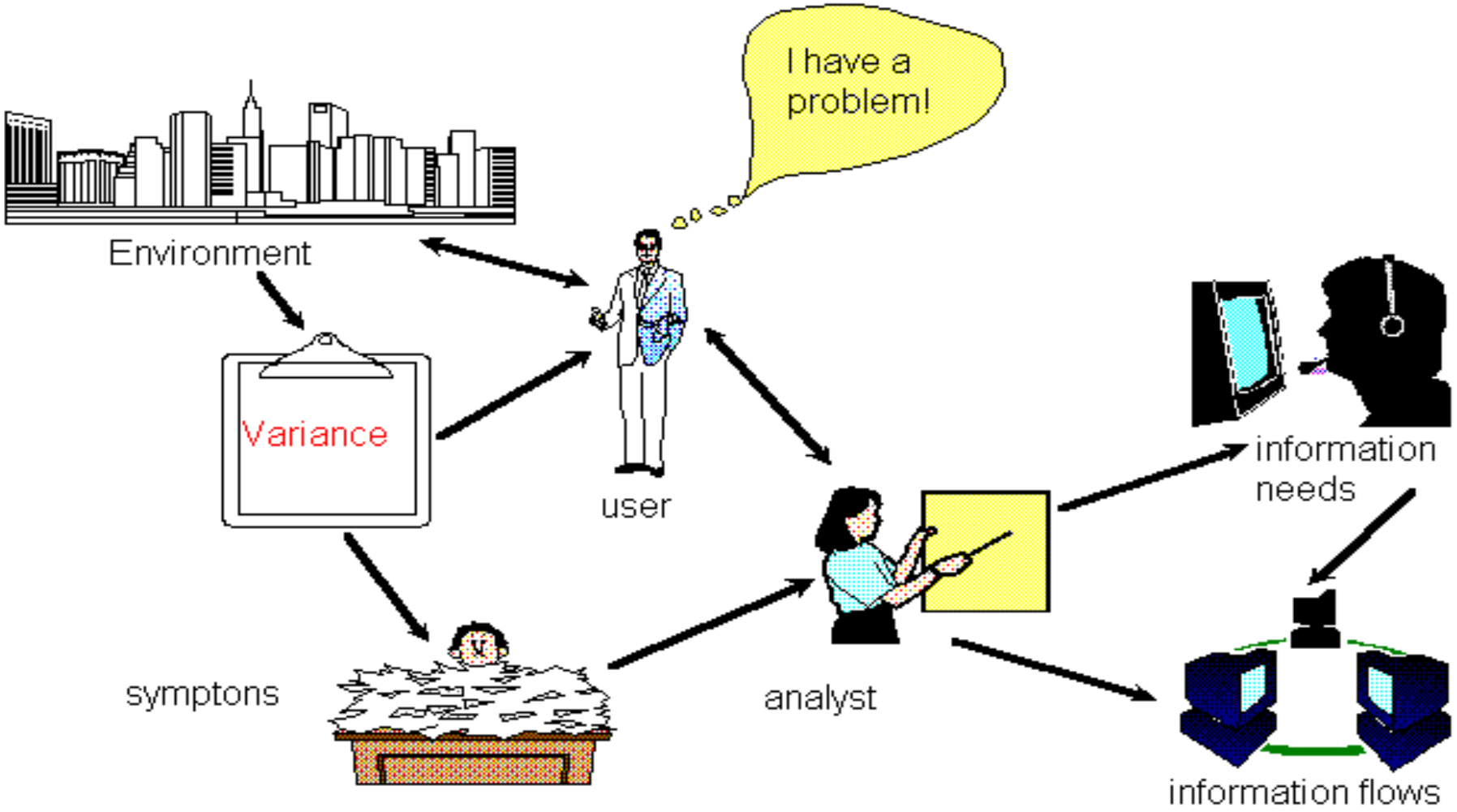
Stakeholders …

- cont..

Systems Analysts

Bridge the communication gap that exists between **nontechnical** and **technical** people involved with building systems.

Responsible for **bridging the gap** between IT and the Business



Systems Development Environment…

- Stakeholders cont..

Systems Analysts

What does a systems analyst do?

- Identify the problem
- Analyze and understand the problem
- Identify the solution requirements
- Identify alternative solutions
- Select the Best Alternative

Systems Development Environment…

- **<u>Alternative Solutions</u>**

Standalone system

Network Sytem

Web Based System

Stakeholders

System Designer

Design the system to meet the users Requirements

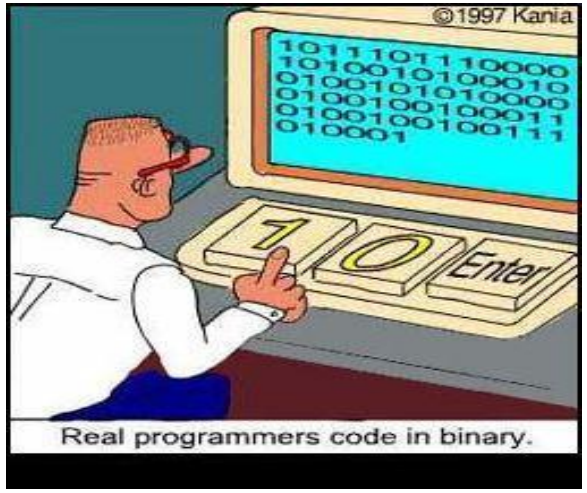
Eg. Database Administrators
Web Architects
Network Architect

- Stakeholders cont..

System Designer

Design the **databases**, **inputs, outputs, screens**, **networks**, and **programs** that will meet the system users requirements.

- Stakeholders cont..



System Builders

Construct, test and deliver the Information System.

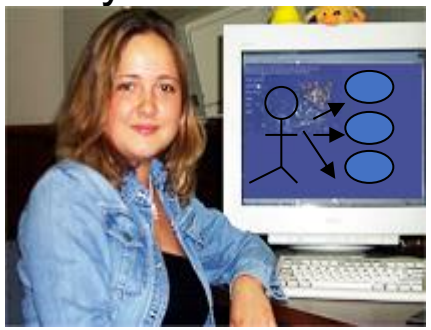
Systems Development Environment…

Information System Development Teams usually consists of the **representatives of these Stakeholders**

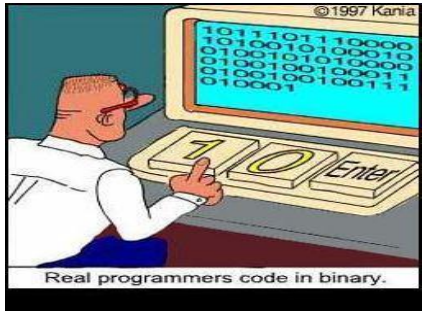
System User

System Owner

System Analysts



System Designer



System Builders

Large scale projects – Interface designers, Database designers, QA staff, Interface programmers, Application programmers etc.

System Development Life Cycle (SDLC)

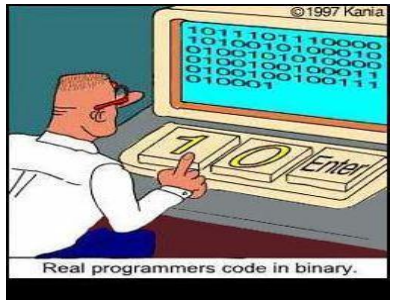
Problem Definition

(Scope Definition)

Requirement Analysis

System Design

System Development



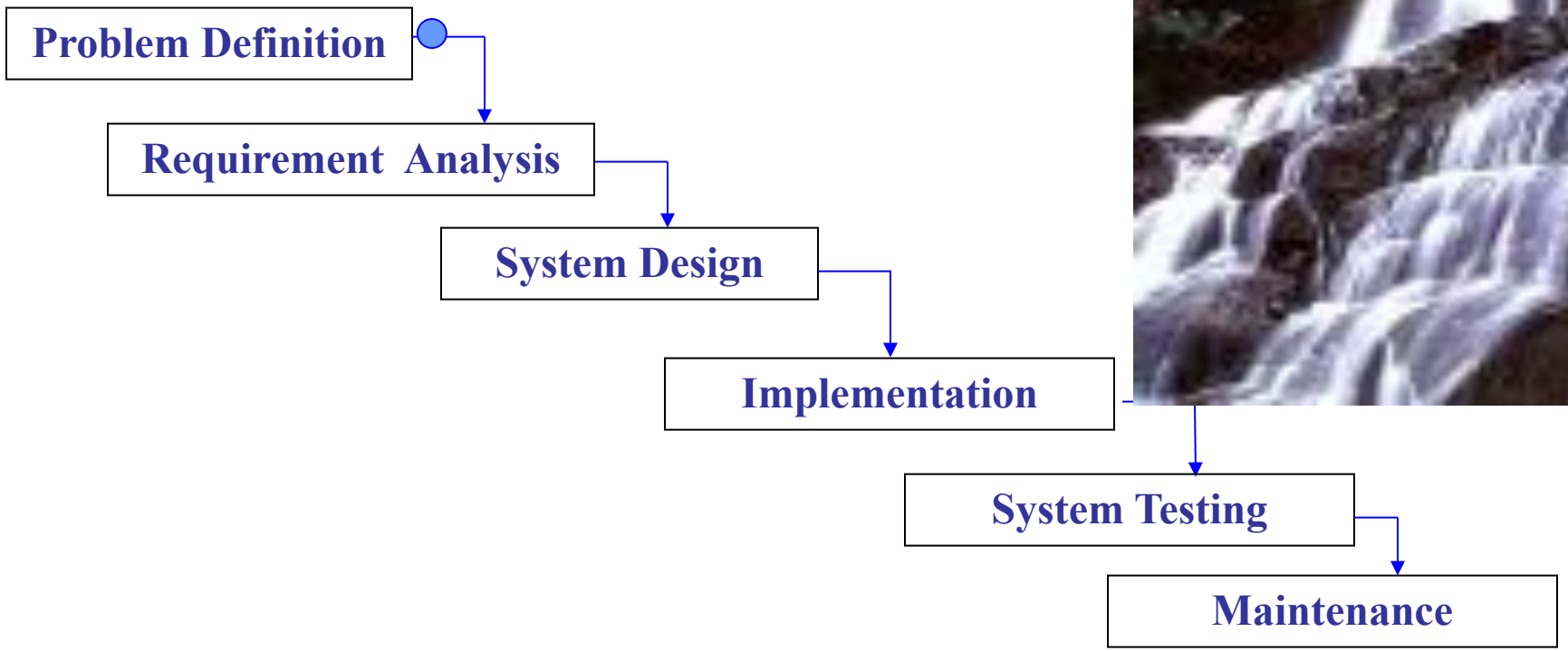
System Testing

Maintenance

System Development Life Cycle (SDLC)

- A systematic approach to software development
- Composed of several phases,
  - **Problem Definition (Systems initiation, Scope definition**) identifies and defines a need for the new system
  - **Systems Analysis** - analyzes the information needs of the end users
  - **Systems Design** - creates a blueprint for the design with the necessary specifications for the hardware, software, people and data resources
  - **Systems Implementation**- writes programs for the final system
  - **Systems Testing** - evaluates the system's actual functionality in relation to expected or intended functionality.
  - **Maintenance** – keeping the system up to date with the changes in the organization and ensuring it meets the goals of the organization

Waterfall Model



A Software Process Model

The linear model groups development activities

into a sequence of consecutive phases.

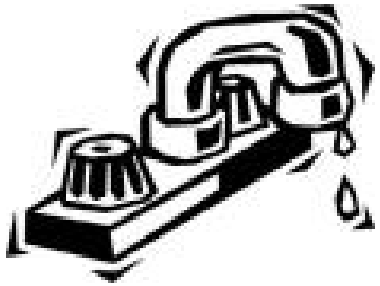
Problem Definition (Scope Definition, System Initiation)

Project goals

Provides a broad statement of user requirements in users terms, or what the users expect the system to do

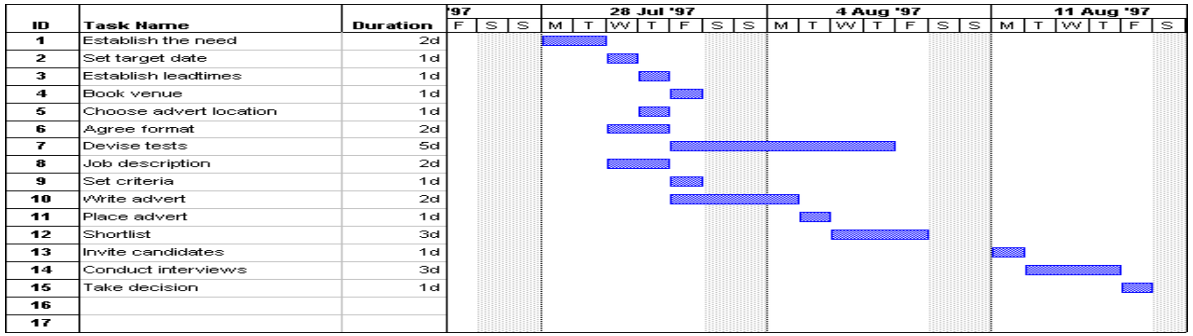
Project limits

project bounds are set during this phase. Defines what part of the system are handled by the project and what parts are to remain same.



Project Schedule

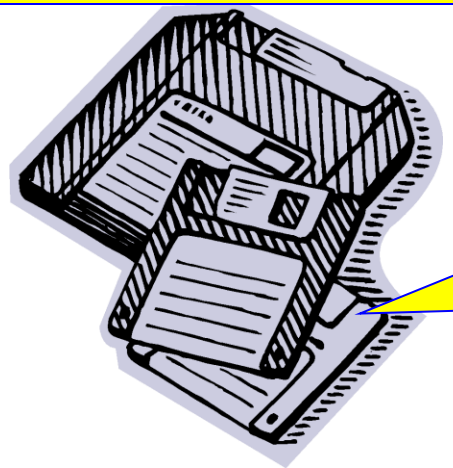
Specify the resources to be made available for the project (*resource limits*).



Requirement Analysis (System Analysis)

How the current system works and what it does

Producing a model of what the new system will need



Producing a high-level description of the system specifying the business requirements and priorities

System Design

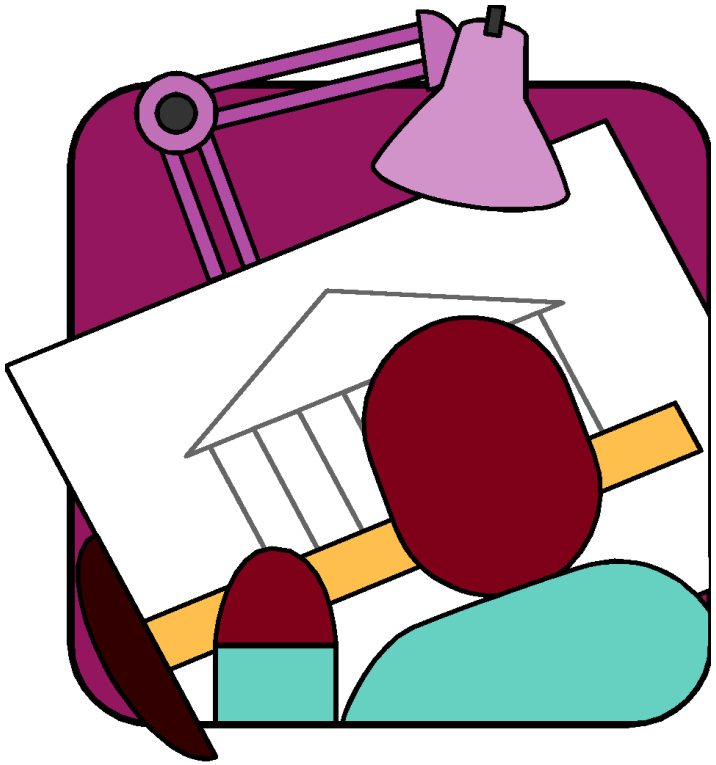
- Describes **how the system will operate**, in terms of the hardware, software, and network infrastructure
- **Produces a specification of a technical, computer based solution** for the business requirements identified in Systems Analysis

Analysts

Design

System Design

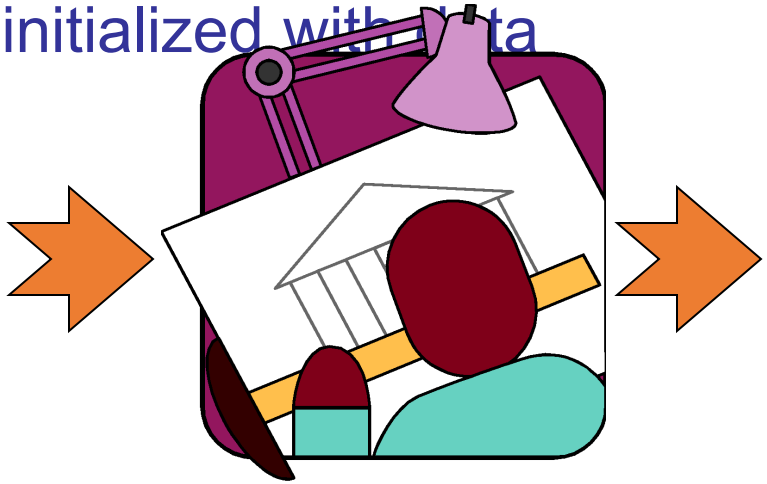
- Things to be done:
  - Explore alternative technical solutions
  - Identify the best solution
  - Develop technical models and specifications to implement
    - Required databases
    - Programs
    - User interfaces
    - Networks etc.



Design

System Development (System Implementation)

- Individual system modules/components are built and tested
- Tools are used to build the system
- User interfaces are developed and tried by users
- Database is initialized with data



Analysis **Design** **Development**

System testing

- Test and evaluate results, and
- the system ready to be delivered to the user/client.

Recognized as an important part of quality assurance.

Maintenance

- Eliminate errors in the system during its working life.
- Fixing any bugs and problems found by users
- Tune the system to any variations in its working environment

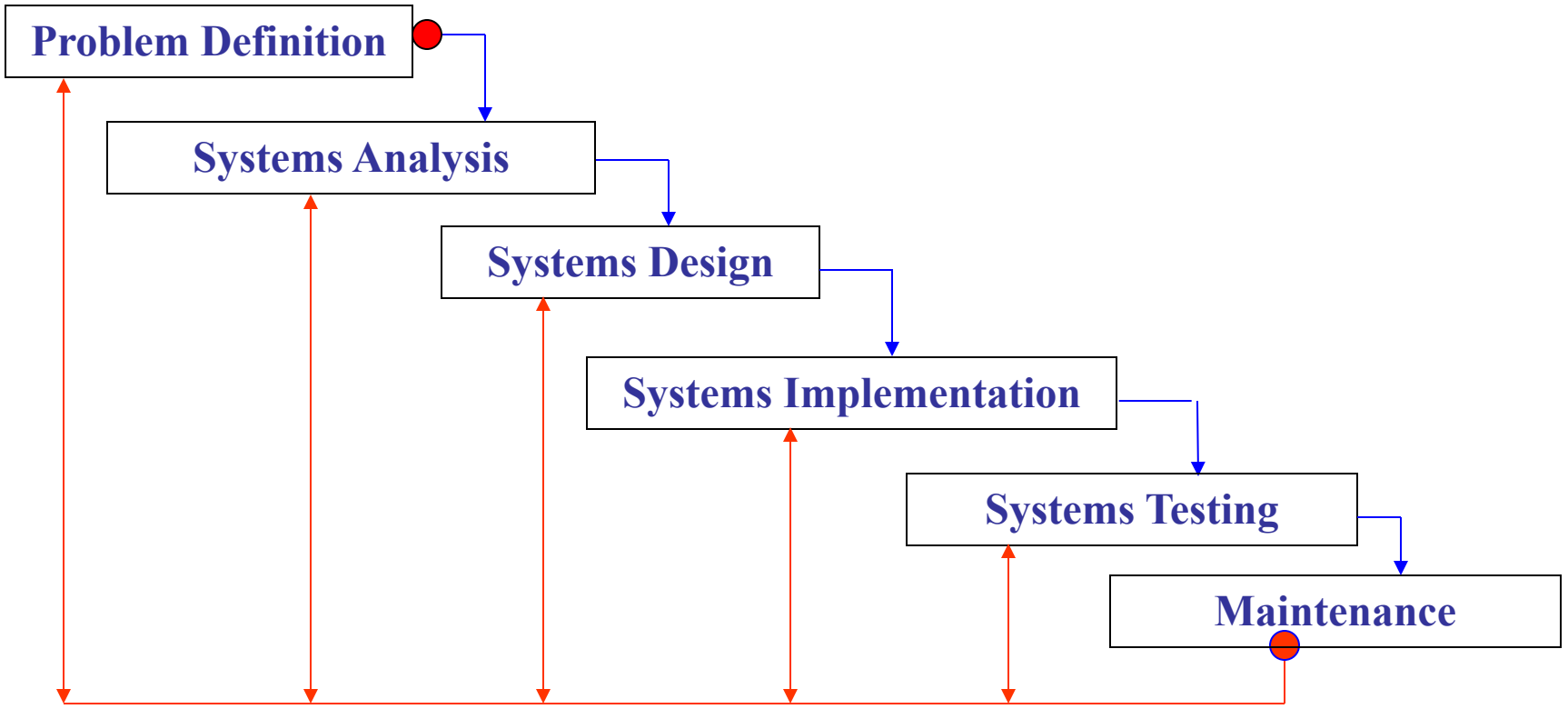
Problems with the waterfall development approach

- ✓ It has a rigid design
- ✓ It has a top-down procedure
- ✓ One phase must be completed before the next phase starts
- ✓ Time consuming
- ✓ Real projects rarely follow the sequential flow that the model proposes.
- ✓ At the beginning of most projects there is a great deal of uncertainty about requirements and goals - difficult for customers to identify these criteria on a detailed level. The waterfall model does not accommodate this natural uncertainty very well.

Criticisms fall into the following categories: cont…

- ✓ Assumptions made in the early phases no longer hold
- ✓ Some of the early work is incomplete
- ✓ Something was overlooked or not completely understood.

Modified Waterfall development approach



Modified Waterfall Development Approach

- Uses the same phases as the pure waterfall development approach.
- Allow some of the stages to overlap, such as the system analysis phase and the systems design stage
- Make it possible to integrate feedback from one phase to another
- Incorporate prototyping.
- Implementation of easy areas do not need to wait for the harder ones
- Progress is more difficult to track